

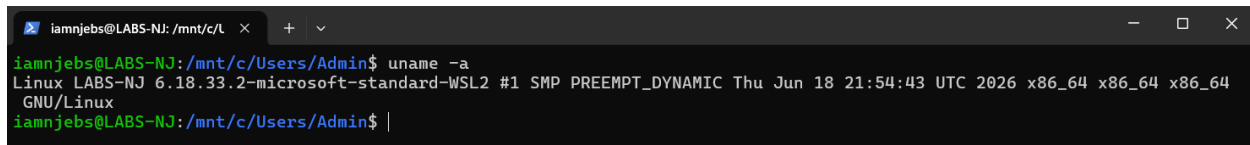
Day 05

LAB 01 – Linux Environment Discovery

Before changing a system, learn to Identify the environment you are working on

Show kernel & system information

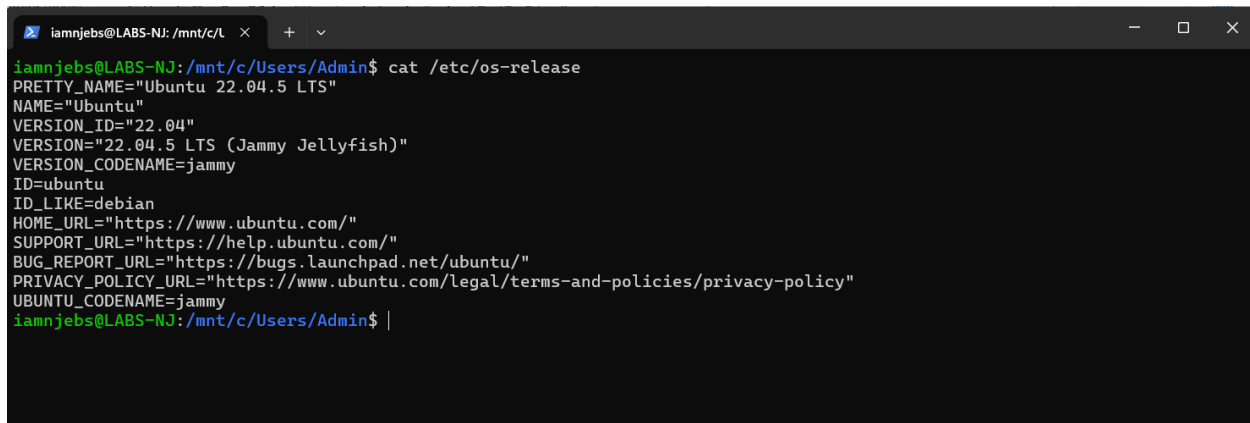
- ***uname -a***



```
iamnjebs@LABS-NJ: /mnt/c/L  X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ uname -a
Linux LABS-NJ 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64
GNU/Linux
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ |
```

Show system OS information

- ***cat /etc/os/-release***



```
iamnjebs@LABS-NJ: /mnt/c/L  X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ |
```

Show system hostname

- ***hostname***



```
iamnjebs@LABS-NJ: /mnt/c/L  X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ hostname
LABS-NJ
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Show network interfaces & IP address

- **ip addr**

```
iamnjebs@LABS-NJ: /mnt/c/L x + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet :::::/32 brd ::::: scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::::: scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:95:3b:71 brd ff:ff:ff:ff:ff:ff
    inet 172.30.250.11/20 brd 172.30.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 ::::: scope link
        valid_lft forever preferred_lft forever
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Show disk usage (human readable)

- **df -h**

```
iamnjebs@LABS-NJ: /mnt/c/L x + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            7.8G  0  7.8G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            7.8G  4.0K  7.8G  1% /mnt/wsl
drivers         476G  261G  216G  55% /usr/lib/wsl/drivers
/dev/sdd        1007G  1.8G  954G  1% /
none            7.8G  44K  7.8G  1% /mnt/wslg
none            7.8G  0  7.8G  0% /usr/lib/wsl/lib
rootfs          7.8G  2.8M  7.8G  1% /init
none            7.8G  520K  7.8G  1% /run
none            7.8G  0  7.8G  0% /run/lock
none            7.8G  0  7.8G  0% /run/shm
none            7.8G  80K  7.8G  1% /mnt/wslg/versions.txt
none            7.8G  80K  7.8G  1% /mnt/wslg/doc
C:\             476G  261G  216G  55% /mnt/c
tmpfs           1.6G  8.0K  1.6G  1% /run/user/1000
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Show Available memory (human readable)

- **free -h**

```
iamnjebs@LABS-NJ: /mnt/c/L x + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ free -h
              total        used        free      shared  buff/cache   available
Mem:          15Gi         392Mi        14Gi         3.0Mi        147Mi        14Gi
Swap:         4.0Gi           0B         4.0Gi
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

? Engineering question: Why should an engineer inspect a system before making changes?

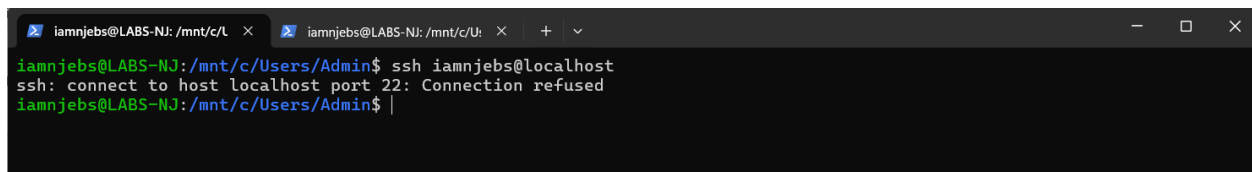
Answers:

- to make sure you know their system capacity, for example system name, OS, CPU, Memory and what limitations the system has.
- This helps engineers to identify what, where they need to make/ modify system changes example directory folder, files, script etc. etc.
- And this prevents downtime, avoids unnecessary costs and protects data security.

LAB 02 SSH Access

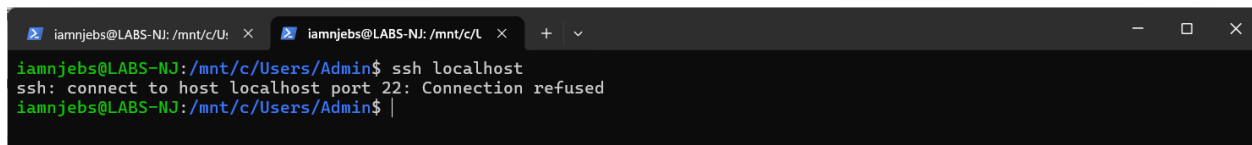
Tried to ssh to localhost

- ***ssh iamnjebs@localhost***

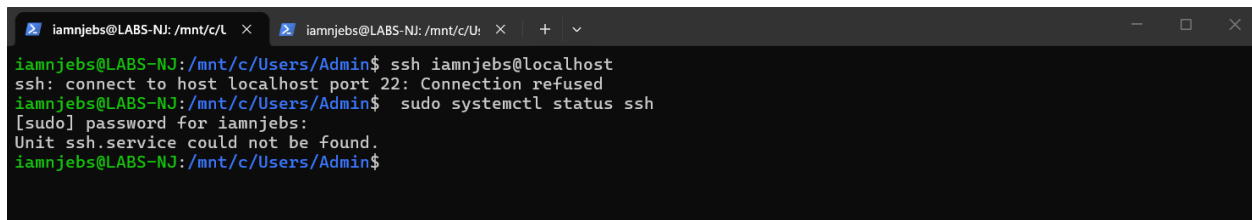


```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ ssh iamnjebs@localhost
ssh: connect to host localhost port 22: Connection refused
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ |
```

- ***ssh localhost***



```
iamnjebs@LABS-NJ: /mnt/c/U:  x  iamnjebs@LABS-NJ: /mnt/c/L  x  +  v
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ ssh localhost
ssh: connect to host localhost port 22: Connection refused
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ |
```



```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ ssh iamnjebs@localhost
ssh: connect to host localhost port 22: Connection refused
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ sudo systemctl status ssh
[sudo] password for iamnjebs:
Unit ssh.service could not be found.
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$
```

Install OpenSSH Server

- ***sudo apt update***

```
iamnjebs@LABS-NJ: /mnt/c/L X iamnjebs@LABS-NJ: /mnt/c/U: X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3855 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3579 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1285 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1052 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [236 kB]
Fetched 10.4 MB in 25s (410 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
26 packages can be upgraded. Run 'apt list --upgradable' to see them.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

- ***sudo apt install openssh-server -y***

```
iamnjebs@LABS-NJ: /mnt/c/L X iamnjebs@LABS-NJ: /mnt/c/U: X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libwrap0 ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  libwrap0 ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 5 newly installed, 0 to remove and 26 not upgraded.
Need to get 800 kB of archives.
After this operation, 6162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.17 [38.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwrap0 amd64 7.6.q-31build2 [47.9 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.17 [436 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.3 [268 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1 [10.1 kB]
Fetched 800 kB in 3s (309 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 42717 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a8.9p1-3ubuntu0.17_amd64.deb ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.17) ...
Selecting previously unselected package libwrap0:amd64.
Preparing to unpack .../libwrap0_7.6.q-31build2_amd64.deb ...
Unpacking libwrap0:amd64 (7.6.q-31build2) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a8.9p1-3ubuntu0.17_amd64.deb ...
Unpacking openssh-server (1:8.9p1-3ubuntu0.17) ...
```

```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
Unpacking openssh-server (1:8.9p1-3ubuntu0.17) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.3-2ubuntu0.3_all.deb ...
Unpacking ncurses-term (6.3-2ubuntu0.3) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu1) ...
Setting up openssh-sftp-server (1:8.9p1-3ubuntu0.17) ...
Setting up ssh-import-id (5.11-0ubuntu1) ...
Setting up libwrap0:amd64 (7.6.q-31build2) ...
Setting up ncurses-term (6.3-2ubuntu0.3) ...
Setting up openssh-server (1:8.9p1-3ubuntu0.17) ...

Creating config file /etc/ssh/sshd_config with new version
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:WlrBenJ+Y4YybhTa+jlDx0xQi4bB0+ZhpDburZyFt4k root@LABS-NJ (RSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:mmh5yXIa7XfSVlaoHpYN9YZQc41ZXPsbcb0shx4nLbs root@LABS-NJ (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:mS7aQ4YkDaZn/jke2BQmytnVOPrxAf4mtmLMxX2T+AU root@LABS-NJ (ED25519)
Created symlink /etc/systemd/system/ssh.service → /lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /lib/systemd/system/ssh.service.
rescue-ssh.target is a disabled or a static unit, not starting it.
ssh.socket is a disabled or a static unit, not starting it.
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$
```

Start & Enable the SSH Service

- ***sudo apt enable --now ssh***

```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$
```

Verify SSH Service Status

- ***sudo systemctl status ssh***

```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-09-24 21:43:32 SAST; 6min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 2423 (sshd)
      Tasks: 1 (limit: 19060)
     Memory: 1.7M
        CPU: 40ms
    CGroup: /system.slice/ssh.service
            └─2423 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 24 21:43:32 LABS-NJ systemd[1]: Starting OpenBSD Secure Shell server...
Sep 24 21:43:32 LABS-NJ sshd[2423]: Server listening on 0.0.0.0 port 22.
Sep 24 21:43:32 LABS-NJ sshd[2423]: Server listening on :: port 22.
Sep 24 21:43:32 LABS-NJ systemd[1]: Started OpenBSD Secure Shell server.
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$
```

Connect to Localhost

- **ssh iamnjebs@localhost**
- **ssh iamnjebs@172.30.250.11**

```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-09-24 21:43:32 SAST; 6min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 2423 (sshd)
    Tasks: 1 (limit: 19060)
   Memory: 1.7M
      CPU: 40ms
   CGroup: /system.slice/ssh.service
           └─2423 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 24 21:43:32 LABS-NJ systemd[1]: Starting OpenBSD Secure Shell server...
Sep 24 21:43:32 LABS-NJ sshd[2423]: Server listening on 0.0.0.0 port 22.
Sep 24 21:43:32 LABS-NJ sshd[2423]: Server listening on :: port 22.
Sep 24 21:43:32 LABS-NJ systemd[1]: Started OpenBSD Secure Shell server.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ ssh iamnjebs@172.30.250.11
The authenticity of host '172.30.250.11 (172.30.250.11)' can't be established.
ED25519 key fingerprint is SHA256:mS7aQ4YkDaZn/jke2BQmytnVOPrxAf4mtmLMxX2T+AU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ ssh iamnjebs@172.30.250.11
The authenticity of host '172.30.250.11 (172.30.250.11)' can't be established.
ED25519 key fingerprint is SHA256:mS7aQ4YkDaZn/jke2BQmytnVOPrxAf4mtmLMxX2T+AU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.30.250.11' (ED25519) to the list of known hosts.
ssh_dispatch_run_fatal: Connection to 172.30.250.11 port 22: Broken pipe
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ |
```

```
iamnjebs@LABS-NJ: ~  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
Sep 24 21:43:32 LABS-NJ sshd[2423]: Server listening on :: port 22.
Sep 24 21:43:32 LABS-NJ systemd[1]: Started OpenBSD Secure Shell server.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ ssh iamnjebs@172.30.250.11
The authenticity of host '172.30.250.11 (172.30.250.11)' can't be established.
ED25519 key fingerprint is SHA256:mS7aQ4YkDaZn/jke2BQmytnVOPrxAf4mtmLMxX2T+AU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.30.250.11' (ED25519) to the list of known hosts.
ssh_dispatch_run_fatal: Connection to 172.30.250.11 port 22: Broken pipe
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ ssh iamnjebs@172.30.250.11
iamnjebs@172.30.250.11's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

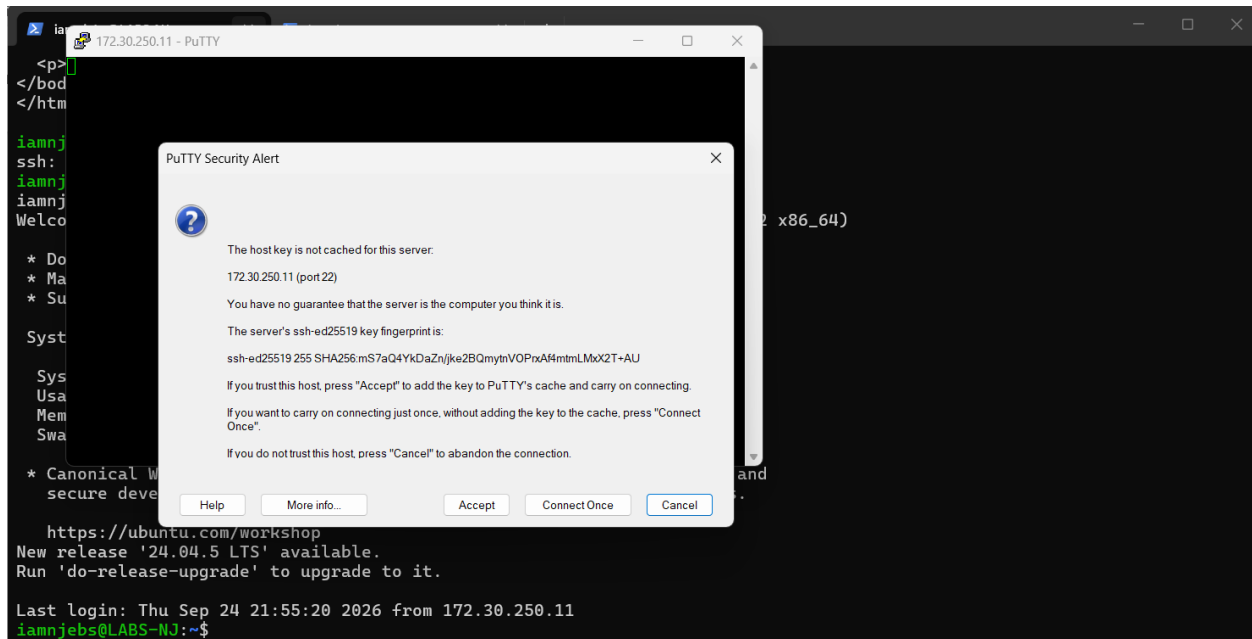
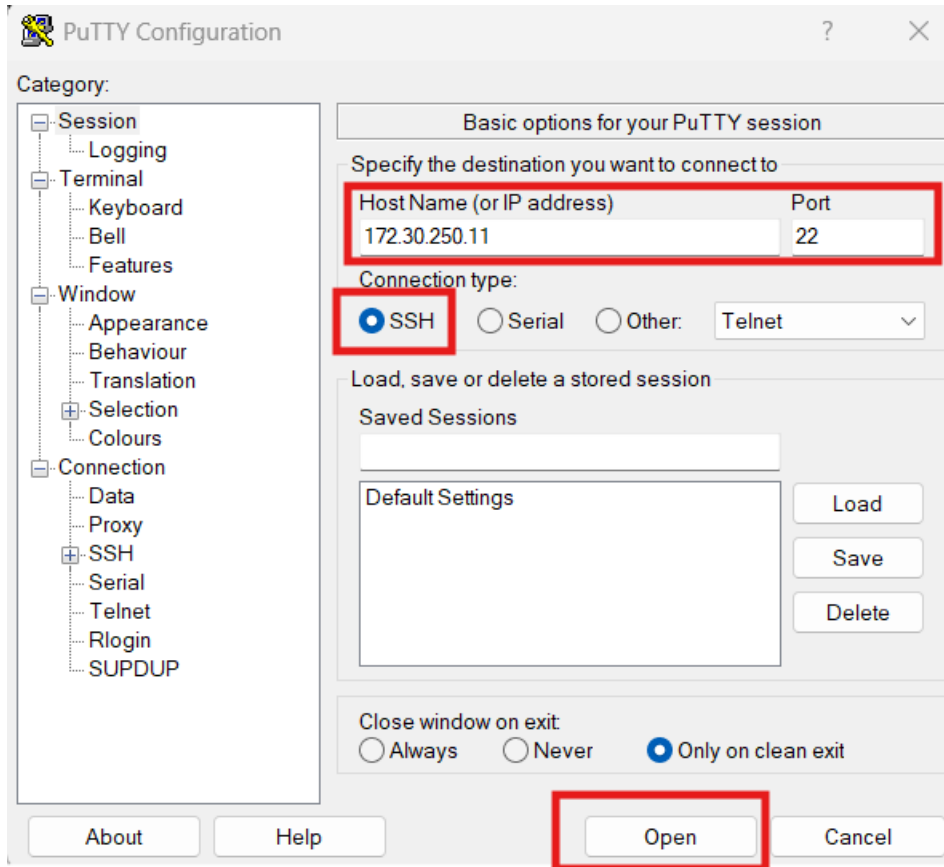
System information as of Thu Sep 24 21:55:20 SAST 2026

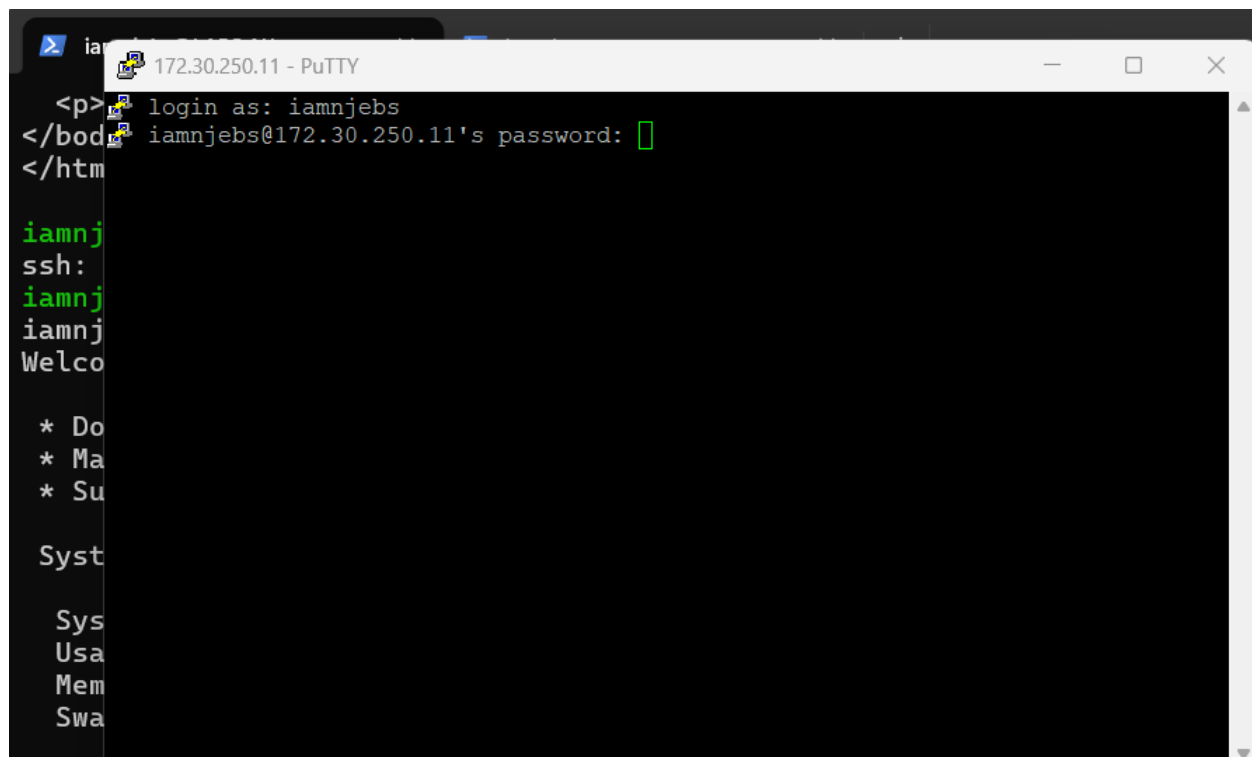
System load:  0.01          Processes:      47
Usage of /:   0.2% of 1006.85GB  Users logged in: 1
Memory usage: 3%             IPv4 address for eth0: 172.30.250.11
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop
Last login: Thu Sep 24 12:47:17 2026
iamnjebs@LABS-NJ:~$
```

Connecting using Putty





A screenshot of a PuTTY terminal window titled "172.30.250.11 - PuTTY". The terminal shows an SSH login session. The prompt is "<p> login as: iamnjebs". The user enters "iamnjebs@172.30.250.11's password:" followed by a green cursor. The prompt changes to "iamnj". The user enters "ssh:". The prompt changes to "iamnj". The user enters "iamnj". The prompt changes to "Welco". The user enters "* Do". The prompt changes to "* Ma". The user enters "* Su". The prompt changes to "Syst". The user enters "Sys". The prompt changes to "Usa". The user enters "Mem". The prompt changes to "Swa".

```
<p> login as: iamnjebs
iamnjebs@172.30.250.11's password: 
iamnj
ssh:
iamnj
iamnj
Welco

* Do
* Ma
* Su

Syst

Sys
Usa
Mem
Swa
```

Inside Putty: output

```
iamnjeks@LABS-NJ: ~  
login as: iamnjeks  
iamnjeks@172.30.250.11's password:  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Thu Sep 24 22:00:21 SAST 2026  
  
System load:  0.0                Processes:            54  
Usage of /:   0.2% of 1006.85GB  Users logged in:     1  
Memory usage: 3%                IPv4 address for eth0: 172.30.250.11  
Swap usage:   0%  
  
* Canonical Workshop gives developers fast, composable, reproducible, and  
  secure developer environments that are perfect for agentic workflows.  
  
  https://ubuntu.com/workshop  
New release '24.04.5 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Thu Sep 24 21:58:17 2026 from 172.30.250.11  
iamnjeks@LABS-NJ:~$ whoami  
iamnjeks  
iamnjeks@LABS-NJ:~$
```

SSH Service Disable

- ***sudo systemctl disable --now ssh***

```
iamnjeks@LABS-NJ:~$ sudo systemctl disable --now ssh  
[sudo] password for iamnjeks:  
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install disable ssh  
Removed /etc/systemd/system/ssh.service.  
Removed /etc/systemd/system/multi-user.target.wants/ssh.service.
```

Verify SSH Service Status

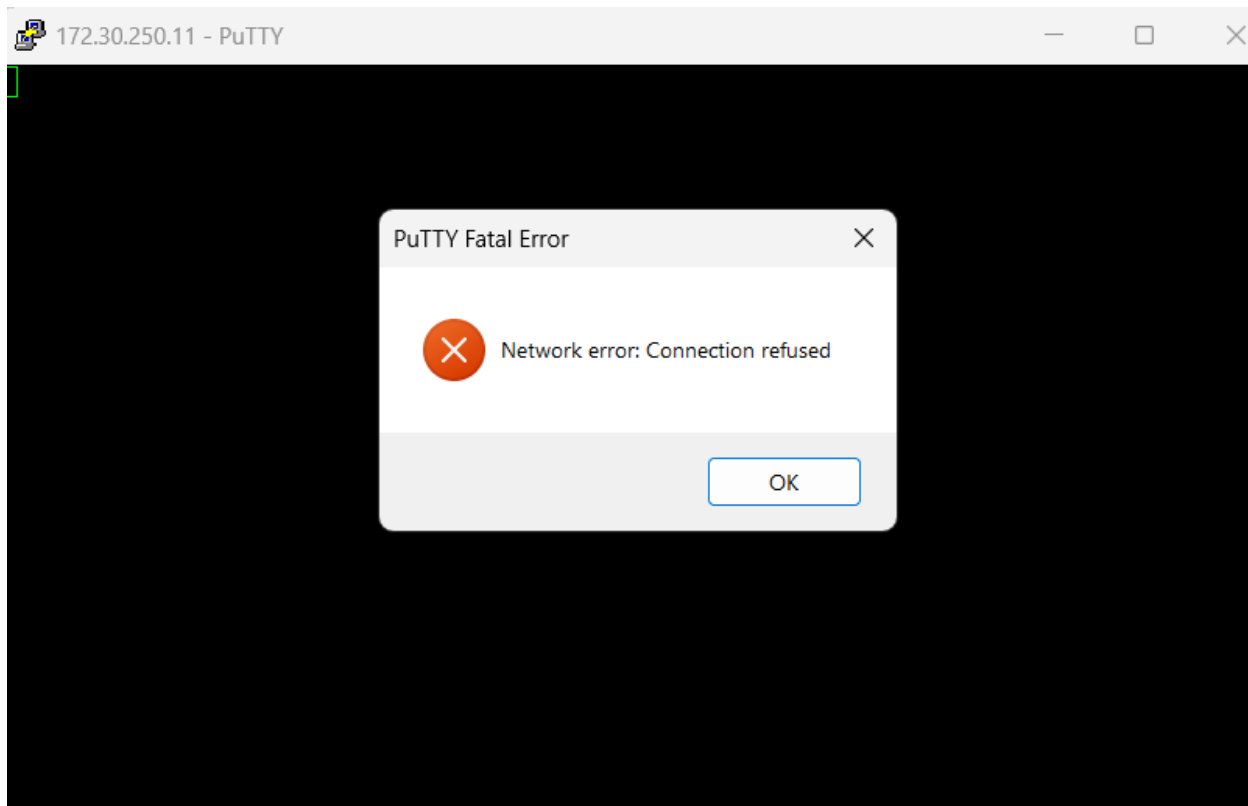
- ***sudo systemctl status ssh***

```
iamnjebs@LABS-NJ:~$ sudo systemctl status ssh
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; vendor preset: enabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)

Sep 24 21:55:20 LABS-NJ sshd[2631]: Accepted password for iajnjebs from 172.30.250.11 port 57676 ssh2
Sep 24 21:55:20 LABS-NJ sshd[2631]: pam_unix(sshd:session): session opened for user iajnjebs(uid=1000) by (uid=0)
Sep 24 21:58:17 LABS-NJ sshd[2715]: Accepted password for iajnjebs from 172.30.250.11 port 52818 ssh2
Sep 24 21:58:17 LABS-NJ sshd[2715]: pam_unix(sshd:session): session opened for user iajnjebs(uid=1000) by (uid=0)
Sep 24 22:00:20 LABS-NJ sshd[2773]: Accepted password for iajnjebs from 172.30.240.1 port 49232 ssh2
Sep 24 22:00:20 LABS-NJ sshd[2773]: pam_unix(sshd:session): session opened for user iajnjebs(uid=1000) by (uid=0)
Sep 24 22:03:29 LABS-NJ sshd[2423]: Received signal 15; terminating.
Sep 24 22:03:29 LABS-NJ systemd[1]: Stopping OpenBSD Secure Shell server...
Sep 24 22:03:29 LABS-NJ systemd[1]: ssh.service: Deactivated successfully.
Sep 24 22:03:29 LABS-NJ systemd[1]: Stopped OpenBSD Secure Shell server.
iamnjebs@LABS-NJ:~$
```

Putty to confirm ssh is disabled

- **connection refused**



LAB 03 – Install Nginx

Update the package Information:

- ***sudo apt update***

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3855 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3579 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [571 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1285 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [324 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [501 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6390 kB]
88% [11 Packages 4921 kB/6390 kB 77%] 291 kB/s 9s|
```

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3855 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3579 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [571 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1285 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [324 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [501 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6390 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [1229 kB]
Fetched 18.0 MB in 44s (407 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
23 packages can be upgraded. Run 'apt list --upgradable' to see them.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Install Nginx

- ***sudo apt install nginx -y***

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.18.0-6ubuntu14.21).
0 upgraded, 0 newly installed, 0 to remove and 23 not upgraded.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Check the service:

- ***sudo systemctl status nginx***

```
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-09-22 20:05:54 SAST; 59s ago
     Docs: man:nginx(8)
   Process: 16439 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 16440 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 16541 (nginx)
      Tasks: 13 (limit: 19060)
     Memory: 25.2M
        CPU: 66ms
    CGroup: /system.slice/nginx.service
            └─16541 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─16543 "nginx: worker process"
                └─16544 "nginx: worker process"
                  └─16545 "nginx: worker process"
                    └─16546 "nginx: worker process"
                      └─16547 "nginx: worker process"
                        └─16548 "nginx: worker process"
                          └─16549 "nginx: worker process"
                            └─16550 "nginx: worker process"
                              └─16551 "nginx: worker process"
                                └─16552 "nginx: worker process"
                                  └─16553 "nginx: worker process"
                                    └─16554 "nginx: worker process"

Sep 22 20:05:54 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 22 20:05:54 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$
```

If necessary, start & enable nginx:

- ***sudo systemctl status nginx***
- ***sudo systemctl enable nginx***

```
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl start nginx
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nginx
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$
```

Verify

- ***curl localhost***

Stop Service

- ***sudo systemctl stop***

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl stop nginx
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Check Service Status

- ***sudo systemctl status nginx***

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl stop nginx
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl status nginx
○ nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: inactive (dead) since Thu 2026-09-24 15:56:31 SAST; 48s ago
     Docs: man:nginx(8)
   Process: 1147 ExecStop=/sbin/start-stop-daemon --quiet --stop --retry QUIT/5 --pidfile /run/nginx.pid (code=exited,
 Main PID: 256 (code=exited, status=0/SUCCESS)
    CPU: 192ms

Sep 24 12:47:15 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 24 12:47:15 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 24 15:56:31 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 24 15:56:31 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 24 15:56:31 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
lines 1-13/13 (END)
```

Start Service

- ***sudo systemctl start nginx***

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl start nginx
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Check Service Status

- ***sudo systemctl status nginx***

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl start nginx
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-09-24 15:58:47 SAST; 30s ago
     Docs: man:nginx(8)
   Process: 1168 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 1169 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Main PID: 1170 (nginx)
    Tasks: 13 (limit: 19060)
   Memory: 20.1M
      CPU: 48ms
   CGroup: /system.slice/nginx.service
           └─1170 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1171 "nginx: worker process"
             └─1172 "nginx: worker process"
             └─1173 "nginx: worker process"
             └─1174 "nginx: worker process"
             └─1175 "nginx: worker process"
             └─1176 "nginx: worker process"
             └─1177 "nginx: worker process"
             └─1178 "nginx: worker process"
             └─1179 "nginx: worker process"
             └─1180 "nginx: worker process"
             └─1181 "nginx: worker process"
             └─1182 "nginx: worker process"

Sep 24 15:58:47 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 24 15:58:47 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
lines 1-27/27 (END)
```

Think about it

What changed between active (running) and inactive (dead)?

💡 Engineering principle: A service can be installed on a machine without currently running.

Answer: System status

LAB 05 Build Your Own Web Page

Web Page HTML Code:

```
<!DOCTYPE html>

<html>

<head>

<title>MZ-UCA Cloud Engineering</title>
```

```
</head>

<body>

  <h1>MZ-UCA Cloud Engineering Lab</h1>

  <p>Linux + Nginx Web Server</p>

  <p>Engineer: Njabulo Khumalo</p>

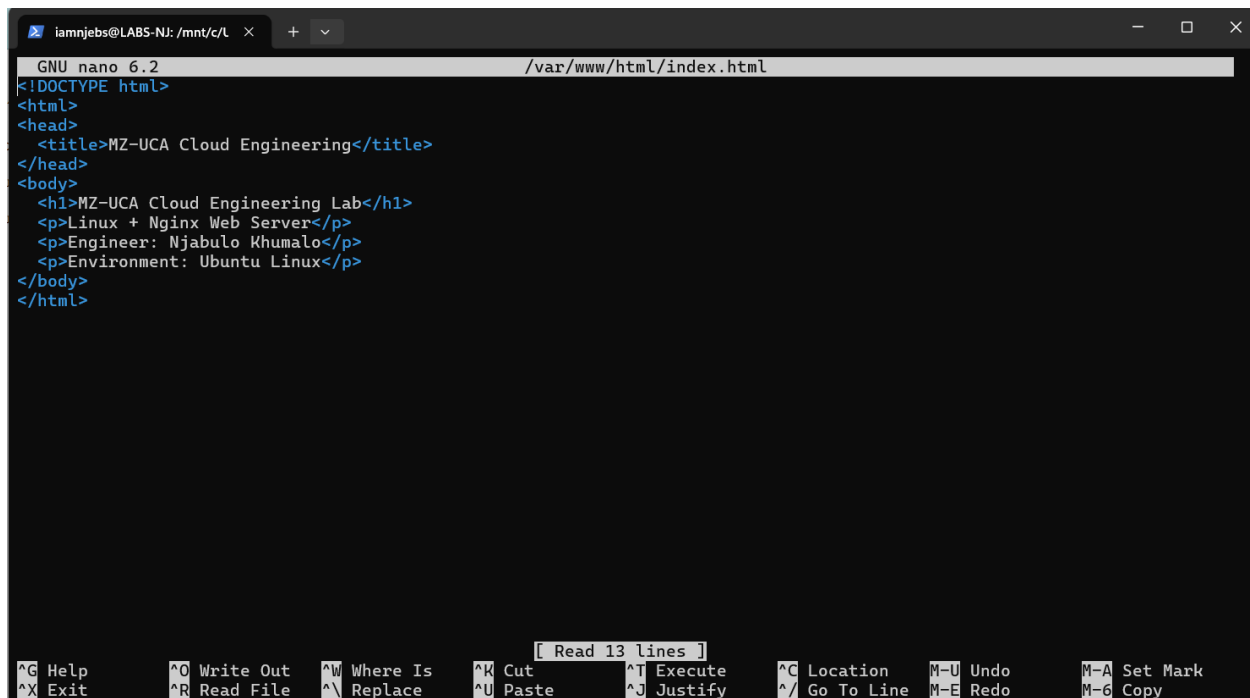
  <p>Environment: Ubuntu Linux</p>

</body>

</html>
```

Open Nano File & Paste the html code.

- ***sudo nano /var/www/html/index.html***



```
iamnjebs@LAB5-NJ: /mnt/c/L x + v
GNU nano 6.2 /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>
[ Read 13 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo     M-6 Copy
```

Save & exit

- ***ctrl + o + enter & ctrl + o + enter***

```
iamnjebs@LABS-NJ: /mnt/c/L × + v
=16550,fd=6),("nginx",pid=16549,fd=6),("nginx",pid=16548,fd=6),("nginx",pid=16547,fd=6),("nginx",pid=16546,fd=6),("nginx",pid=16545,fd=6),("nginx",pid=16544,fd=6),("nginx",pid=16543,fd=6),("nginx",pid=16541,fd=6))
tcp LISTEN 0 1000 10.255.255.254:53 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
users:(("systemd-resolve",pid=2383,fd=14))
tcp LISTEN 0 511 [::]:80 [::]:*
users:(("nginx",pid=16554,fd=7),("nginx",pid=16553,fd=7),("nginx",pid=16552,fd=7),("nginx",pid=16551,fd=7),("nginx",pid=16550,fd=7),("nginx",pid=16549,fd=7),("nginx",pid=16548,fd=7),("nginx",pid=16547,fd=7),("nginx",pid=16546,fd=7),("nginx",pid=16545,fd=7),("nginx",pid=16544,fd=7),("nginx",pid=16543,fd=7),("nginx",pid=16541,fd=7))
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo nano /var/www/html/index.html

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ |
```

Open a browser and search for your webpage.

- ***http://172.30.250.11/***



MZ-UCA Cloud Engineering Lab

Linux + Nginx Web Server

Engineer: Njabulo Khumalo

Environment: Ubuntu Linux

LAB 06 Understand Ports

Check Listening Services

- **`sudo ss -tulpn`**

```
iamnjebs@LABS-NJ: /mnt/c/L  Windows PowerShell

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ sudo ss -tulpn
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port
Process
udp        UNCONN     0           0           127.0.0.53%lo:53        0.0.0.0:*
users:((("systemd-resolve",pid=2383,fd=13))
udp        UNCONN     0           0           10.255.255.254:53       0.0.0.0:*
udp        UNCONN     0           0           127.0.0.1:323          0.0.0.0:*
udp        UNCONN     0           0           [::1]:323             [::]:*
tcp        LISTEN     0           511         0.0.0.0:80             0.0.0.0:*
users:((("nginx",pid=16554,fd=6),("nginx",pid=16553,fd=6),("nginx",pid=16552,fd=6),("nginx",pid=16551,fd=6),("nginx",pid=16550,fd=6),("nginx",pid=16549,fd=6),("nginx",pid=16548,fd=6),("nginx",pid=16547,fd=6),("nginx",pid=16546,fd=6),("nginx",pid=16545,fd=6),("nginx",pid=16544,fd=6),("nginx",pid=16543,fd=6),("nginx",pid=16541,fd=6))
tcp        LISTEN     0           1000        10.255.255.254:53       0.0.0.0:*
tcp        LISTEN     0           4096        127.0.0.53%lo:53        0.0.0.0:*
users:((("systemd-resolve",pid=2383,fd=14))
tcp        LISTEN     0           511         [::]:80               [::]:*
users:((("nginx",pid=16554,fd=7),("nginx",pid=16553,fd=7),("nginx",pid=16552,fd=7),("nginx",pid=16551,fd=7),("nginx",pid=16550,fd=7),("nginx",pid=16549,fd=7),("nginx",pid=16548,fd=7),("nginx",pid=16547,fd=7),("nginx",pid=16546,fd=7),("nginx",pid=16545,fd=7),("nginx",pid=16544,fd=7),("nginx",pid=16543,fd=7),("nginx",pid=16541,fd=7))
iamnjebs@LABS-NJ: /mnt/c/Users/Admin$ |
```

Listening Ports:

- Protocol (TCP/UDP) Port **:53 DNS** used for name resolution
- Protocol (TCP) Port **:80 HTTP** used default communication port for unencrypted traffic on the web.

NB: **UNCONN** (Unconnected) ports read to receive incoming packets

- These ports are open & listening for traffic (**:53 DNS** & **:323 IMMP** (Internet Message Mapping Protocol))

Show a specific port: 80

- **`sudo ss -tulpn | grep :80`**

```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /mnt/c/U:  x  +  v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ss -tulpn | grep :80
[sudo] password for iamnjebs:
tcp LISTEN 0 511 0.0.0.0:80 0.0.0.0:* users:(("nginx",pid=1182,fd=6),("nginx",pid=1181,fd=6),
("nginx",pid=1180,fd=6),("nginx",pid=1179,fd=6),("nginx",pid=1178,fd=6),("nginx",pid=1177,fd=6),("nginx",pid=1176,fd=6),
("nginx",pid=1175,fd=6),("nginx",pid=1174,fd=6),("nginx",pid=1173,fd=6),("nginx",pid=1172,fd=6),("nginx",pid=1171,fd=6),
("nginx",pid=1170,fd=6))
tcp LISTEN 0 511 [::]:80 [::]:* users:(("nginx",pid=1182,fd=7),("nginx",pid=1181,fd=7),
("nginx",pid=1180,fd=7),("nginx",pid=1179,fd=7),("nginx",pid=1178,fd=7),("nginx",pid=1177,fd=7),("nginx",pid=1176,fd=7),
("nginx",pid=1175,fd=7),("nginx",pid=1174,fd=7),("nginx",pid=1173,fd=7),("nginx",pid=1172,fd=7),("nginx",pid=1171,fd=7),
("nginx",pid=1170,fd=7))
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

LAB 07 Configure the Firewall

Check for ufw & enable ufw

- ***sudo ufw status***
- ***sudo ufw enable***

```
iamnjebs@LABS-NJ: /mnt/c/L  x  +  v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ufw status
Status: inactive
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ufw enable
Firewall is active and enabled on system startup
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ufw status
Status: active
```

Allow SSH

- ***sudo ufw allow ssh***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ufw allow ssh
Rule added
Rule added (v6)
```

Allow HTTP

- ***sudo ufw allow 80/tcp***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Check ufw status

- ***sudo ufw status***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
```

LAB 08 Test Connectivity

Verify web server responds local

- ***curl localhost***

```
iamnjebs@LABS-NJ:/mnt/c/L x + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>MZ-UCA Cloud Engineering</title>
</head>
<body>
<h1>MZ-UCA Cloud Engineering Lab</h1>
<p>Linux + Nginx Web Server</p>
<p>Engineer: Njabulo Khumalo</p>
<p>Environment: Ubuntu Linux</p>
</body>
</html>

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ |
```

Identify network (IP & Interfaces)

- ***ip addr***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
    inet6 ::: scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:95:39:d1 brd ff:ff:ff:ff:ff:ff
    inet 172.30.250.11/20 brd 172.30.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 ::: scope link
        valid_lft forever preferred_lft forever
```

Verify your server using local server actual IP

- ***curl http://172.30.250.11***

```
iamnjebs@LABS-NJ: /mnt/c/L  X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl http://172.30.250.11
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Test on the Browser (Chrome)

- ***https://172.30.250.11/***



MZ-UCA Cloud Engineering Lab

Linux + Nginx Web Server

Engineer: Njabulo Khumalo

Environment: Ubuntu Linux

LAB 09 Access Logs

Check Logs

- ***sudo tail -f /var/log/nginx/access.log***

```
iamnjebs@LABS-NJ: /mnt/c/L  ×  +  ▾
curl: (6) Could not resolve host: http
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl http://172.30.250.11
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo tail -f /var/log/nginx/access.log
127.0.0.1 - - [22/Sep/2026:20:10:47 +0200] "GET / HTTP/1.1" 200 612 "-" "curl/7.81.0"
172.30.240.1 - - [22/Sep/2026:20:44:06 +0200] "GET / HTTP/1.1" 200 192 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:20:44:06 +0200] "GET /favicon.ico HTTP/1.1" 404 197 "http://172.30.250.11/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:20:55:27 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:21:02:09 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
127.0.0.1 - - [22/Sep/2026:21:05:25 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:21:24:39 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:21:24:57 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
172.30.250.11 - - [22/Sep/2026:21:27:13 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
172.30.250.11 - - [22/Sep/2026:21:30:16 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
```

To escape

- **control + C + Enter**

```
iamnjebs@LABS-NJ: /mnt/c/L  ×  +  ▾
172.30.240.1 - - [22/Sep/2026:20:44:06 +0200] "GET /favicon.ico HTTP/1.1" 404 197 "http://172.30.250.11/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:20:55:27 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:21:02:09 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
127.0.0.1 - - [22/Sep/2026:21:05:25 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:21:24:39 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:21:24:57 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
172.30.250.11 - - [22/Sep/2026:21:27:13 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
172.30.250.11 - - [22/Sep/2026:21:30:16 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"

^C
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo journalctl -u nginx
Sep 22 20:05:54 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 22 20:05:54 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

LAB 09 Access Log

Run and leave terminal running:

- ***sudo tail -f /var/log/nginx/access.log***

```
iamnjebs@LABS-NJ: /mnt/c/L X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo tail -f /var/log/nginx/access.log
127.0.0.1 - - [22/Sep/2026:21:24:39 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:21:24:57 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
172.30.250.11 - - [22/Sep/2026:21:27:13 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
172.30.250.11 - - [22/Sep/2026:21:30:16 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:21:32:52 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
172.30.240.1 - - [22/Sep/2026:21:43:01 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:21:43 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:22:24 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:32 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:33 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:54 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:54 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:55 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:55 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:56 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:58 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
```

Observation:

- When I run (***sudo tail -f /var/log/nginx/access.log***) it shows logs and when I refresh the webpage (localhost) more logs are generated.

New Logs generated when running

- ***curl localhost***

```
iamnjebs@LABS-NJ: /mnt/c/L X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo tail -f /var/log/nginx/access.log
172.30.240.1 - - [22/Sep/2026:22:41:32 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:33 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:54 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:54 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:55 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:55 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:56 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
172.30.240.1 - - [22/Sep/2026:22:41:58 +0200] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36"
127.0.0.1 - - [22/Sep/2026:22:44:20 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:22:44:54 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"
127.0.0.1 - - [22/Sep/2026:22:46:22 +0200] "GET / HTTP/1.1" 200 248 "-" "curl/7.81.0"

iamnjebs@LABS-NJ: /mnt/c/L X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Observation:

- When I ***curl localhost*** new Logs are generated

LAB 10 Error Logs

Application errors investigation

- ***sudo tail -f /var/log/nginx/error.log***
- ***sudo journalctl -u nginx***

```
iamnjebs@LABS-NJ: /mnt/c/L X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo tail -f /var/log/nginx/error.log
2026/09/22 20:05:55 [notice] 16541#16541: using inherited sockets from "6;7;"
^C
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo journalctl -u nginx
Sep 22 20:05:54 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 22 20:05:54 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Admin> wsl
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo tail -f /var/log/nginx/error.log
[sudo] password for iamenjebs:
2026/09/22 20:05:55 [notice] 16541#16541: using inherited sockets from "6;7;"
^C
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo journalctl -u nginx
sudo: journalctl -u: command not found
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo journalctl -u nginx
Sep 22 20:05:54 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 22 20:05:54 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 22 23:09:28 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 22 23:09:28 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 22 23:09:28 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
-- Boot f0fde9e5209a4f688a74d686cac60275 --
Sep 23 20:18:01 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 23 20:18:01 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ |
```

Questions

What is the difference between `access.log` and `error.log`? What type of information might you expect in each?

Answer:

- **`access.log`** are activity records (**web page visits**) and **`error.log`** are problem reports (**web page/server issues**)

LAB 11 - Break The Web Server

Stop Nginx Services

- **`sudo systemctl stop nginx`**

```
iamnjebs@LABS-NJ: /mnt/c/L X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl stop nginx
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
curl: (7) Failed to connect to localhost port 80 after 0 ms: Connection refused
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl status nginx
○ nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: inactive (dead) since Wed 2026-09-23 21:33:51 SAST; 1min 50s ago
     Docs: man:nginx(8)
   Process: 232 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 265 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 1653 ExecStop=/sbin/start-stop-daemon --quiet --stop --retry QUIT/5 --pidfile /run/nginx.pid (code=exited, status=0/SUCCESS)
  Main PID: 276 (code=exited, status=0/SUCCESS)
    CPU: 225ms

Sep 23 20:18:01 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 23 20:18:01 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 23 21:33:51 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
lines 1-15/15 (END)
```

Test

- **curl localhost**

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
curl: (7) Failed to connect to localhost port 80 after 0 ms: Connection refused
```

Investigate

- **sudo systemctl status nginx**

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl status nginx
○ nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: inactive (dead) since Wed 2026-09-23 21:33:51 SAST; 1min 50s ago
     Docs: man:nginx(8)
   Process: 232 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 265 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 1653 ExecStop=/sbin/start-stop-daemon --quiet --stop --retry QUIT/5 --pidfile /run/nginx.pid (code=exited, status=0/SUCCESS)
  Main PID: 276 (code=exited, status=0/SUCCESS)
    CPU: 225ms

Sep 23 20:18:01 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 23 20:18:01 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 23 21:33:51 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
lines 1-15/15 (END)
```

Run logs

- **sudo journalctl -u nginx**

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo journalctl -u nginx
Sep 22 20:05:54 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 22 20:05:54 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 22 23:09:28 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 22 23:09:28 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 22 23:09:28 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
-- Boot f0fde9e5209a4f688a74d686cac60275 --
Sep 23 20:18:01 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 23 20:18:01 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 23 21:33:51 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

The webpage is unavailable because of the **nginx service we deactivated**

Recover the Nginx Service

- `sudo systemctl start nginx`

```
iamnjebs@LABS-NJ: /mnt/c/L  X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ udo systemctl start nginx
Command 'udo' not found, but can be installed with:
sudo apt install udo
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl start nginx
```

Typo error ***udo*** instead of ***sudo***

Test again

- ***curl localhost***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ |
```

LAB 12 - Troubleshooting Challenge

1. Check Service

- ***sudo systemctl status nginx***

```
iamnjeb@LABS-NJ: /mnt/c/L  +  -  x
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-09-23 21:37:58 SAST; 4min 17s ago
     Docs: man:nginx(8)
   Process: 1703 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 1704 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1705 (nginx)
    Tasks: 13 (limit: 19060)
   Memory: 21.0M
      CPU: 67ms
   CGroup: /system.slice/nginx.service
           └─1705 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1706 "nginx: worker process"
               └─1707 "nginx: worker process"
                 └─1708 "nginx: worker process"
                   └─1709 "nginx: worker process"
                     └─1710 "nginx: worker process"
                       └─1711 "nginx: worker process"
                         └─1712 "nginx: worker process"
                           └─1713 "nginx: worker process"
                             └─1714 "nginx: worker process"
                               └─1715 "nginx: worker process"
                                 └─1716 "nginx: worker process"
                                   └─1717 "nginx: worker process"

Sep 23 21:37:58 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 23 21:37:58 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Lines 1-27/27 (END)
```

2. Port

- *sudo ss -tulpn*

```
iamnjeb@LABS-NJ: /mnt/c/L  +  -  x
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$ sudo ss -tulpn
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port
Process
udp        UNCONN     0           0           127.0.0.53%lo:53        0.0.0.0:*
users:(("systemd-resolve",pid=107,fd=13))
udp        UNCONN     0           0           10.255.255.254:53      0.0.0.0:*
udp        UNCONN     0           0           127.0.0.1:323         0.0.0.0:*
udp        UNCONN     0           0           [::]:323             [::]:*
tcp        LISTEN     0          4096         127.0.0.53%lo:53      0.0.0.0:*
users:(("systemd-resolve",pid=107,fd=14))
tcp        LISTEN     0          1000        10.255.255.254:53     0.0.0.0:*
tcp        LISTEN     0           511         0.0.0.0:80           0.0.0.0:*
users:(("nginx",pid=1717,fd=6),("nginx",pid=1716,fd=6),("nginx",pid=1715,fd=6),("nginx",pid=1714,fd=6),("nginx",pid=1713,fd=6),("nginx",pid=1712,fd=6),("nginx",pid=1711,fd=6),("nginx",pid=1710,fd=6),("nginx",pid=1709,fd=6),("nginx",pid=1708,fd=6),("nginx",pid=1707,fd=6),("nginx",pid=1706,fd=6),("nginx",pid=1705,fd=6))
tcp        LISTEN     0           511         [::]:80             [::]:*
users:(("nginx",pid=1717,fd=7),("nginx",pid=1716,fd=7),("nginx",pid=1715,fd=7),("nginx",pid=1714,fd=7),("nginx",pid=1713,fd=7),("nginx",pid=1712,fd=7),("nginx",pid=1711,fd=7),("nginx",pid=1710,fd=7),("nginx",pid=1709,fd=7),("nginx",pid=1708,fd=7),("nginx",pid=1707,fd=7),("nginx",pid=1706,fd=7),("nginx",pid=1705,fd=7))
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$ sudo ss -tulpn | grep :80
tcp        LISTEN     0           511         0.0.0.0:80           0.0.0.0:* users:(("nginx",pid=1717,fd=6),("nginx",pid=1716,fd=6),("nginx",pid=1715,fd=6),("nginx",pid=1714,fd=6),("nginx",pid=1713,fd=6),("nginx",pid=1712,fd=6),("nginx",pid=1711,fd=6),("nginx",pid=1710,fd=6),("nginx",pid=1709,fd=6),("nginx",pid=1708,fd=6),("nginx",pid=1707,fd=6),("nginx",pid=1706,fd=6),("nginx",pid=1705,fd=6))
tcp        LISTEN     0           511         [::]:80             [::]:* users:(("nginx",pid=1717,fd=7),("nginx",pid=1716,fd=7),("nginx",pid=1715,fd=7),("nginx",pid=1714,fd=7),("nginx",pid=1713,fd=7),("nginx",pid=1712,fd=7),("nginx",pid=1711,fd=7),("nginx",pid=1710,fd=7),("nginx",pid=1709,fd=7),("nginx",pid=1708,fd=7),("nginx",pid=1707,fd=7),("nginx",pid=1706,fd=7),("nginx",pid=1705,fd=7))
iamnjeb@LABS-NJ:/mnt/c/Users/Admin$ |
```

Check specific port :80

- ***sudo ss -tulpin | grep :80***

```
iamnjebs@LABS-NJ: /mnt/c/L x + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ss -tulpn | grep :80
tcp LISTEN 0 511 0.0.0.0:80 0.0.0.0:* users:(("nginx",pid=1717,fd=6),("nginx",pid=1716,fd=6),
("nginx",pid=1715,fd=6),("nginx",pid=1714,fd=6),("nginx",pid=1713,fd=6),("nginx",pid=1712,fd=6),("nginx",pid=1711,fd=6),
("nginx",pid=1710,fd=6),("nginx",pid=1709,fd=6),("nginx",pid=1708,fd=6),("nginx",pid=1707,fd=6),("nginx",pid=1706,fd=6),
("nginx",pid=1705,fd=6))
tcp LISTEN 0 511 [::]:80 [::]:* users:(("nginx",pid=1717,fd=7),("nginx",pid=1716,fd=7),
("nginx",pid=1715,fd=7),("nginx",pid=1714,fd=7),("nginx",pid=1713,fd=7),("nginx",pid=1712,fd=7),("nginx",pid=1711,fd=7),
("nginx",pid=1710,fd=7),("nginx",pid=1709,fd=7),("nginx",pid=1708,fd=7),("nginx",pid=1707,fd=7),("nginx",pid=1706,fd=7),
("nginx",pid=1705,fd=7))
```

3. Network

- ***ip addr***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet  ::::1/1 scope global lo
        valid_lft forever preferred_lft forever
    inet6  ::1/1 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:95:3d:6d brd ff:ff:ff:ff:ff:ff
    inet 172.30.250.11/20 brd 172.30.250.31 scope global eth0
        valid_lft forever preferred_lft forever
    inet6  fe80::15:5d:95:3d:6d%eth0 scope link
        valid_lft forever preferred_lft forever
```

4. Firewall

- ***sudo ufw status***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

5. Logs

- ***sudo journalctl -u nginx***

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ sudo journalctl -u nginx
Sep 22 20:05:54 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 22 20:05:54 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 22 23:09:28 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 22 23:09:28 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 22 23:09:28 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
-- Boot f0fde9e5209a4f688a74d686cac60275 --
Sep 23 20:18:01 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 23 20:18:01 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopping A high performance web server and a reverse proxy server...
Sep 23 21:33:51 LABS-NJ systemd[1]: nginx.service: Deactivated successfully.
Sep 23 21:33:51 LABS-NJ systemd[1]: Stopped A high performance web server and a reverse proxy server.
Sep 23 21:37:58 LABS-NJ systemd[1]: Starting A high performance web server and a reverse proxy server...
Sep 23 21:37:58 LABS-NJ systemd[1]: Started A high performance web server and a reverse proxy server.
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

6. Configuration

- *curl localhost*

```
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ curl localhost
<!DOCTYPE html>
<html>
<head>
  <title>MZ-UCA Cloud Engineering</title>
</head>
<body>
  <h1>MZ-UCA Cloud Engineering Lab</h1>
  <p>Linux + Nginx Web Server</p>
  <p>Engineer: Njabulo Khumalo</p>
  <p>Environment: Ubuntu Linux</p>
</body>
</html>
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$
```

Problem: The website was down

Evidence: systemctl displayed inactive and curl localhost displayed fail

Root cause: Nginx stopped, port 80 was blocked by the ufw, there were some misconfigurations

Resolution: I restart Nginx, fixed the ufw and correct configurations

Verification: the webpage successfully returned by testing *curl localhost*

LAB 13 - Engineering Evidence

Screenshots Folder

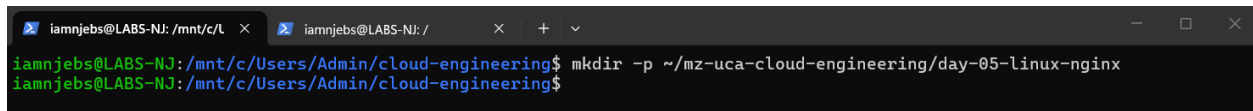
- name: **day-05-screenshots**

```
iamnjebs@LABS-NJ: ~/mz-uc  X + v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ cd ~/mz-uca-cloud-engineering
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering$ ls
day-05-linux-nginx
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering$ cd day-05-linux-nginx
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering/day-05-linux-nginx$ ls
README.md  day-05-screenshots  notes.md
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering/day-05-linux-nginx$ cd day-05-screenshots
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering/day-05-linux-nginx/day-05-screenshots$ ls
'Evidence 02 - Nginx running.png'  'Evidence 05 - Nginx Access Logs.png'
'Evidence 03 - My Custom Webpage.png'  'Evidence 06 Troubleshooting & Resolving Investigation.png'
'Evidence 04 - Firewall Rules.png'  'Evidence 01 - Linux Environment.png'
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering/day-05-linux-nginx/day-05-screenshots$
```

LAB 14 Create Your Lab Notes

Created Folder Outside Cloud-Engineering Folder

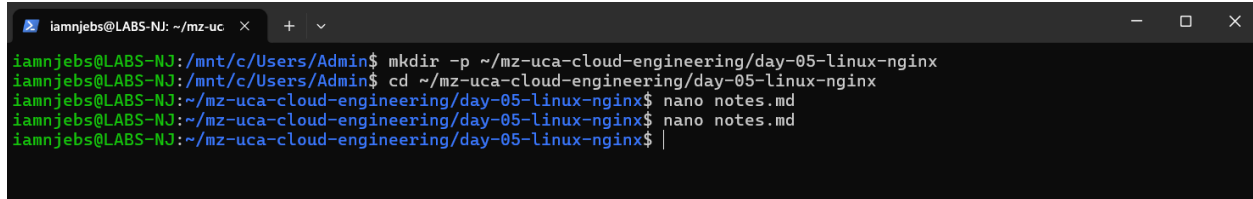
- ***mkdir -p ~/mz-uca-cloud-engineering/day-05-linux-nginx***



```
iamnjebs@LABS-NJ: /mnt/c/L  x  iamnjebs@LABS-NJ: /  x  +  v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin/cloud-engineering$ mkdir -p ~/mz-uca-cloud-engineering/day-05-linux-nginx
iamnjebs@LABS-NJ:/mnt/c/Users/Admin/cloud-engineering$
```

Inside the Folder

- ***cd ~/mz-uca-cloud-engineering/day-05-linux-nginx***



```
iamnjebs@LABS-NJ: ~/mz-uc  x  +  v
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ mkdir -p ~/mz-uca-cloud-engineering/day-05-linux-nginx
iamnjebs@LABS-NJ:/mnt/c/Users/Admin$ cd ~/mz-uca-cloud-engineering/day-05-linux-nginx
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering/day-05-linux-nginx$ nano notes.md
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering/day-05-linux-nginx$ nano notes.md
iamnjebs@LABS-NJ:~/mz-uca-cloud-engineering/day-05-linux-nginx$ |
```

Create a notes file:

- ***nano notes.md***

Paste as shown below save and exit

The screenshot shows a terminal window with the title bar 'iamnjeb@LAB5-NJ: ~/mz-uc'. The terminal is running the GNU nano 6.2 text editor, editing a file named 'notes.md'. The file's content is as follows:

```
# Day 05 - Linux + Nginx Lab
## Environment
Operating System:
Hostname:
IP Address:
## Nginx
Installation:
Service Status:
Web Root:
## Networking
SSH Port:
HTTP Port:
## Firewall
Rules configured:
## Testing
Tests performed:
## Troubleshooting
Failure:
Evidence:
Root Cause:
Resolution:
## Lessons Learned
1.
2.
3.
```

At the bottom of the terminal, a status bar displays '[Read 25 lines]' and a series of keyboard shortcuts for nano editor commands, such as '^G Help', '^O Write Out', '^W Where Is', '^K Cut', '^T Execute', '^C Location', '^U Undo', '^M-A Set Mark', '^X Exit', '^R Read File', '^_ Replace', '^U Paste', '^J Justify', '^/_ Go To Line', '^M-E Redo', and '^M-6 Copy'.

Git Upload

Created another folder for documantion

- mkdir documentation

```
iamnjeb@LABS-NJ: ~/mz-uc
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx$ ls
README.md  day-05-screenshots  notes.md
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx$ mkdir documentation
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx$ ls
README.md  day-05-screenshots  documentation  notes.md
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx$ cd documentation
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx/documentation$ ls
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx/documentation$ cd ..
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx$ mv documentation documentation
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx$ ls
README.md  day-05-screenshots  documentation  notes.md
iamnjeb@LABS-NJ:~/mz-uc-cloud-engineering/day-05-linux-nginx$ |
```

I update/rename the typo error for documentation

- *mv documentantation documentation*

main

1 Branch

0 Tags

Go to file

Code

iamrjebz · screenshots update · 10 hours ago · 3 Commits

day-05-screenshots · screenshots update · 10 hours ago

README.md · Initial commit · yesterday

notes.md · Initial commit · yesterday

README

Labs Completed

- Environment discovery
- SSH access
- Nginx installation
- Custom webpage deployment
- Firewall configuration
- Connectivity testing
- Logs inspection
- Troubleshooting and recovery

Evidence

Screenshots and notes are included in this folder.

About

This repository documents my practical engineering labs for Linux and Nginx, completed as part of the MZ-UCA Cloud Engineering program. It demonstrates environment discovery, secure remote access, web server deployment

📖 README

🔗 Activity

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🔗 0 forks

📁 Report repository

Releases

No releases published

Contributors

iamrjebz